THIN AIR

Four-fifths of the gas that forms the atmosphere, including nearly all its water vapor, is concentrated in the troposphere. The higher you go, the thinner the air gets. If you were to travel just 6 miles (10 km) up, you would not find enough air to keep you alive, which is why mountain climbers and high-altitude skydivers carry their own air supplies.

▲ HIGH JUMP
In 2012, skydiver Felix Baumgartner jumped from a record altitude of 24 miles (39 km). He fell faster than the speed of sound, because there was almost no air to slow him down.

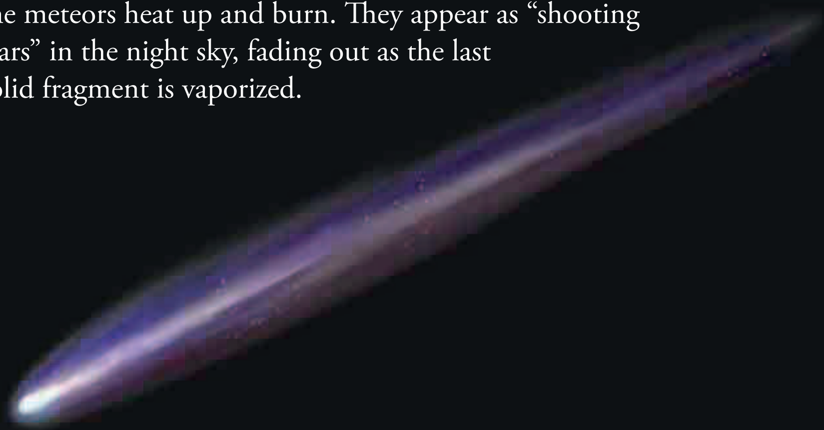
VITAL OXYGEN

Nearly all the oxygen in the atmosphere was created more than two billion years ago by tiny microbes called cyanobacteria. They used energy from sunlight to turn carbon dioxide and water into sugar and oxygen in a process called photosynthesis. Green plants use the same process today, releasing oxygen into the atmosphere.

▲ GREEN LEAVES
Chlorophyll, a green-colored compound used in photosynthesis, gives leaves their color.

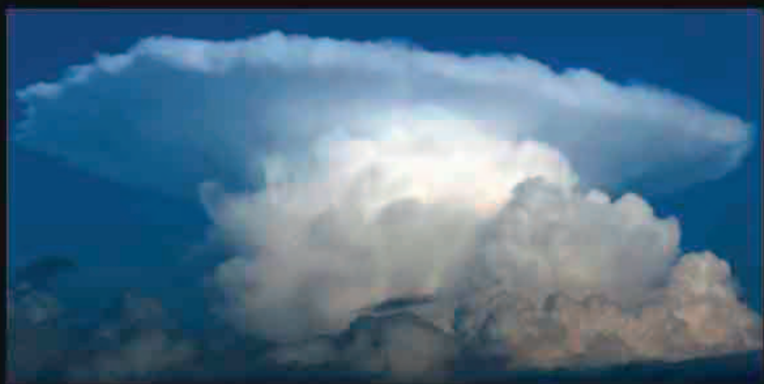
BURN OUT

In the upper atmosphere, the air is very thin. However, there is enough air to slow down small rocky meteors as they hurtle toward Earth's surface. Friction with air molecules makes the meteors heat up and burn. They appear as "shooting stars" in the night sky, fading out as the last solid fragment is vaporized.



TOP OF THE WEATHER

Earth's weather takes place in the troposphere—the lowest layer of the atmosphere. This is because a change in air temperature in the stratosphere stops warm air from rising into it. Instead, the air spreads sideways, taking the weather with it. You can see this happening when big thunderclouds grow tall enough to reach the edge of the stratosphere and flatten out.



◂ BLUE GLOW
Viewed from an orbiting spacecraft, Earth's atmosphere appears blue because gas molecules scatter blue light, while the light of other colors passes straight through.